BE 25: Biophysical Chemistry

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1 Measurement of chemical rates

We have learned how to go from a putative reaction mechanism to differential equations describing the dynamics of concentrations of chemical species using mass action kinetics. Central to this analysis is the determination of the constants of proportionality between the product of concentrations of reactants and the reaction rate. We have come to know these as **rate constants**.

We are going to defer some of the theory behind rate constants, in particular concepts of temperature-dependence and diffusion-limited rates, until we have built the thermodynamic machinery we need to understand these concepts. For now, in this lesson, we will focus on methods for monitoring reactions such that kinetic rate constants may be *empirically* determined.

1.1 Monitoring reactions with light

Optical methods are among the most widely used for monitoring biochemical reactions. I will not go into detail about these methods here, but instead encourage you to read sections 2.6 and 2.7 of Wittrup, Tidor, Hackel, and Sarkar. Below is a brief summary of optical methods.

1.1.1 Scattering

When light impinges upon biomolecules, it can induce oscillations in the molecules. The oscillations result in a re-radiation, which is measured as scattered intensity. It is commonly used in flow cytometry to get a rough estimate of the size and shape of objects.

1.1.2 Absorbance

More than just scattering can happen when biomolecules are exposed to light. If the light has a wavelength (and therefore energy) corresponding to electronic transitions allowable in the molecule, the impinging photons may trigger the electronic transitions, and the photon is absorbed and does not exit the sample. By monitoring how much light gets through a sample, absorbance can be measured. For dilute solutions, the absorbance, given by $A = \ln(I_0/I)$, where I_0 is the intensity of incident light and I is the intensity of transmitted light, is directly proportional to absorbing species concentration, thereby allowing concentration determination.

1.1.3 Fluorescence

This is perhaps the most widely used method for monitoring quantity and location of biochemical species in use today. Some molecules can **fluoresce**. The process of fluorescence involves four steps.

1. **Excitation.** Incident light of wavelength corresponding to an excitation energy kicks the molecule into a higher energy excited electronic state. This happens on a femtosecond time scale.
2. **Excited radiationless transition.** While still in the excited electronic state, the molecule relaxes to a lower vibrational state. No photon is emitted in this transition. The lifetime of this state is on the order of nanoseconds, but can sometimes last as long as a microsecond.
3. **Emission.** The molecule relaxes back down to the ground electronic state. The difference in energy is accounted for by the release (emission) of a photon. Sometimes, a photon is not emitted and the energy is dissipated as heat instead. Emission happens on a femtosecond time scale.
4. **Ground radiationless transition.** As a final relaxation, the molecule drops to a low vibrational state within the ground electronic state. No photon is emitted.

Importantly, the time scale of the processed of fluorescence are much faster than most processes of biological interest. Unfortunately, not many molecules fluoresce, which is why those that do, called **fluorophores**, are often covalently bound to molecules of interest.

The photons emitted by fluorescence are measured by an assortment of detectors, including sophisticated microscopes that also allow for spatial data about the fluorophores.

Quenchers are molecules that, when in contact with an excited fluorophore, accept the energy from the excited state and rapidly convert it to heat, resulting in no photon emission. This process is called **quenching**. The efficiency E_{F} of quenching falls rapidly with the distance between the fluorophore and quencher according to

$$E_{\mathrm{F}} = \frac{1}{1 + (r - r_0)^6}, \tag{1.1}$$

where r is the distance between the fluorophore and quencher and r_0 is the **Förster radius**, named after Theodor Förster, who worked out the relation. As a result of this rapid decay of quenching efficiency with distance, molecules are often constructed with a fluorophore-quencher pair on the same molecule. This allows for experimental monitoring of conformations of biomolecules. Such experiments are called **FRET experiments**, an acronym for fluorescence (or Förster) resonance energy transfer.

1.2 Stopped flow experiments

There are many challenges for measuring the kinetic rate constants of fast reactions, such as those catalyzed by enzymes. One particularly daunting challenge is to completely mix the reactants without letting too much of the reaction proceed. For very slow reactions, one can simply pipette reactants into a cuvette and then measure absorbance, fluorescence, or whatever readout of species concentrations are applicable. But for fast reactions that proceed on time scales of milliseconds, this is very challenging.

In addition to relaxation experiments, described in the next section, **stopped flow experiments** provide a means to rapidly mix components and then quickly monitor optical properties of the reaction mixture over time. A schematic of a stopped flow apparatus is shown in Fig. 1. In this example, purified solutions of enzyme and substrate are added together via syringes into a microfluidic device. The device is designed to induce microscale turbulence immediately after the reactant streams come into contact, which accomplishes rapid mixing. After mixing, the reaction mixture enters a region called the sample cell, which has optically clear walls enabling spectroscopic detection. Typically the detectors can function at very high frequency, often in tens or hundreds of kHz. Once the input streams have mixed and entered the sample cell, the stop switch is triggered to stop the flow, and the reaction is monitored.

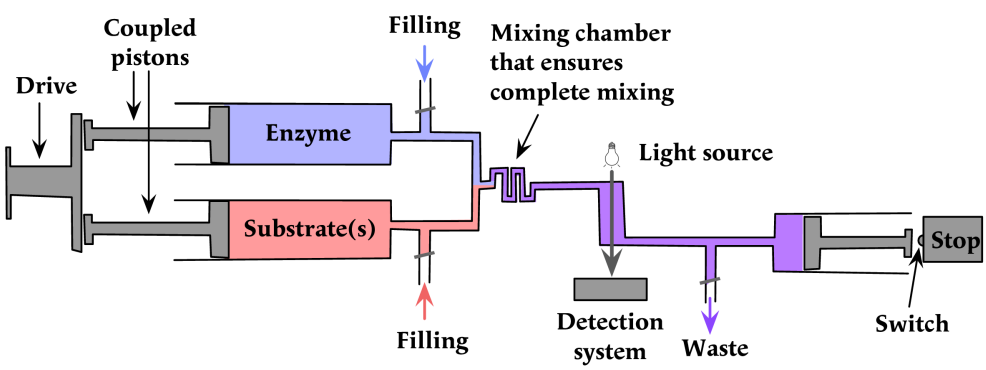

Figure 1: Set up of a stopped flow experiment for measuring enzyme kinetics. Image created by Athel Cornish-Bowden, licensed under a CC-BY-SA 4.0 license.

Ideally, stopped flow apparatuses have very low **dead time**, which is the amount of time that passes between the point when the reactant mixture first comes into contact and when it can be observed. Modern stopped flow devices can have dead times as low as hundreds of nanoseconds.

As an example of how a stopped flow apparatus may work, consider a simple reaction involving the binding of a ligand A to a receptor B that can undergo the

reaction

$$\mathrm{A} + \mathrm{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{AB}. \tag{1.2}$$

The ligand A has a fluorophore attached to it, and the receptor B has a quencher, such that the progress in the reaction can be monitored by loss of fluorescence signal over time. Assuming complete quenching when A is bound, the fluorescence is

$$f = f_0\, c_{\mathrm{A}}, \tag{1.3}$$

where f_0 is a constant of proportionality between the concentration of A and the measured fluorescence. We will call a function that maps chemical concentrations to a measured quantity a **readout function** (my term; I do not know of an "official" term for this). We have previously worked out that if the inlet concentrations of A and B are equal such that $c_{\mathrm{A}}^0 = c_{\mathrm{B}}^0 = c_0$,

$$c_{\mathrm{A}}(t) = \frac{c_0}{1 + c_0\, k\, t}. \tag{1.4}$$

Thus,

$$f(t) = \frac{f_0\, c_0}{1 + c_0\, k\, t} \tag{1.5}$$

Since c_0 is known f_0 and k can be found by performing a regression using this equation on the decay of fluorescence in the sample chamber over time.

1.3 Relaxation experiments

Relaxation experiments are a commonly used method for assessing reaction rates of reversible reactions. In a relaxation experiment, a reaction mixture is allowed to come to a steady state (where all time derivatives vanish). Then, conditions are suddenly and subtly changed and the dynamics observed. This usually involves a jump in temperature, pH, or pressure. Temperature or pH jumps can be made very rapidly using pulsed lasers to raise the temperature or photolyze absorbing molecules, respectively. In any case, the steady state (or equilibrium as we will soon learn about) shifts, since rate constants are in general dependent on temperature, pH, etc.

As an example, consider the reversible binding,

$$\mathrm{A} + \mathrm{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{AB}\,\cdot \tag{1.6}$$

The dynamical equations for this reaction are

$$\frac{\mathrm{d}c_{\mathrm{AB}}}{\mathrm{d}t} = -\frac{\mathrm{d}c_{\mathrm{A}}}{\mathrm{d}t} = -\frac{\mathrm{d}c_{\mathrm{B}}}{\mathrm{d}t} = k_1\, c_{\mathrm{A}}\, c_{\mathrm{B}} - k_{-1}\, c_{\mathrm{AB}}. \tag{1.7}$$

Alternatively, we can write a single differential equation in terms of the extent of reaction, $\phi = c_{\mathrm{AB}} - c^0_{\mathrm{AB}}$ as

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = k_1(c^0_{\mathrm{A}} - \phi)(c^0_{\mathrm{B}} - \phi) - k_{-1}(c^0_{\mathrm{AB}} + \phi). \tag{1.8}$$

We can define the **fractional conversion** ξ as the extent of reaction divided by the total concentration of the limiting reagent (this is a general definition). Assuming A is limiting such that $c^0_{\mathrm{A}} \leq c^0_{\mathrm{B}}$,

$$\xi = \phi / c^0_{\mathrm{A}}. \tag{1.9}$$

Dividing the differential equation by c^0_{A}, and assuming $c^0_{\mathrm{AB}} = 0$, we have

$$\frac{\mathrm{d}\xi}{\mathrm{d}t} = k_1\, c^0_{\mathrm{A}}(1 - \xi)\left(\frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} - \xi\right) - k_{-1}\, \xi. \tag{1.10}$$

We can show that for any set of reversible reactions in a closed system (one in which no material flows in or out) with dynamics governed by mass action that a steady state exists and is unique.[1] We will denote the steady state fractional conversion as ξ_{ss}. It is not important to find it for the discussion here, but it can be solved to be

$$\xi_{\mathrm{ss}} = \frac{1}{2}\left(1 + \frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} + \frac{k_{-1}}{k_1\, c^0_{\mathrm{A}}} - \sqrt{\left(1 + \frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} + \frac{k_{-1}}{k_1\, c^0_{\mathrm{A}}}\right)^2 - 4\,\frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}}}\right). \tag{1.11}$$

Now, let's say we quickly make a perturbation, say by a temperature jump, such that the steady state shifts. Let us assume that immediate after the perturbation the new extent of reaction is $\xi_{\mathrm{ss}} + \delta\xi$, where $\delta\xi$ is a small perturbation. After the temperature jump, the reaction mixture will relax back to steady state according to the above differential equation, or

$$\frac{\mathrm{d}(\xi_{\mathrm{ss}} + \delta\xi)}{\mathrm{d}t} = k_1(1 - \xi_{\mathrm{ss}} - \delta\xi)\left(\frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} - \xi_{\mathrm{ss}} - \delta\xi\right) - k_{-1}(\xi_{\mathrm{ss}} + \delta\xi). \tag{1.12}$$

We can group terms as

$$\begin{aligned}\left[\frac{\mathrm{d}\xi_{\mathrm{ss}}}{\mathrm{d}t}\right] + \frac{\mathrm{d}\delta\xi}{\mathrm{d}t} = &\left[k_1\, c^0_{\mathrm{A}}(1 - \xi_{\mathrm{ss}})\left(\frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} - \xi_{\mathrm{ss}}\right) - k_{-1}\, \xi_{\mathrm{ss}}\right] \\ &- k_1\, c^0_{\mathrm{A}}\left(\left(1 + \frac{c^0_{\mathrm{B}}}{c^0_{\mathrm{A}}} - 2\,\xi_{\mathrm{ss}}\right)\delta\xi - (\delta\xi)^2\right) - k_{-1}\, \delta\xi\end{aligned} \tag{1.13}$$

[1] We will prove this when we talk about equilibrium in the thermodynamics section of the course.

The bracketed terms are zero by the definition of the steady state. Further, if the perturbation is small, the $(\delta\xi)^2$ is negligible compared to those linear in $\delta\xi$. Thus, we have

$$\frac{\mathrm{d}\,\delta\xi}{\mathrm{d}t} = -\left(k_1\, c_{\mathrm{A}}^0 \left(1 + \frac{c_{\mathrm{B}}^0}{c_{\mathrm{A}}^0} - 2\,\xi_{\mathrm{ss}}\right) + k_{-1}\right)\ \delta\xi\,. \tag{1.14}$$

This is a first order linear differential equation with solution

$$\delta\xi\,(t) = \delta\xi_0\,\mathrm{e}^{-t/\tau}, \tag{1.15}$$

with the relaxation time τ given by

$$\tau = \frac{1}{k_1\, c_{\mathrm{A}}^0 \left(1 + \frac{c_{\mathrm{B}}^0}{c_{\mathrm{A}}^0} - 2\,\xi_{\mathrm{ss}}\right) + k_{-1}}. \tag{1.16}$$

It is traditional to write this relaxation time not in terms of the steady state fractional conversion, but in terms of the steady state concentrations, $c_{\mathrm{A}}^{\mathrm{ss}} = c_{\mathrm{A}}^0 - \xi_{\mathrm{ss}}$ and $c_{\mathrm{B}}^{\mathrm{ss}} = c_{\mathrm{B}}^0 - \xi_{\mathrm{ss}}$.

$$\tau = \frac{1}{k_1\,(c_{\mathrm{A}}^{\mathrm{ss}} + c_{\mathrm{B}}^{\mathrm{ss}}) + k_{-1}}. \tag{1.17}$$

To determine the rate constants k_1 and k_{-1}, one can perform experiments for various concentrations of reactants, thereby giving different steady state concentrations. For each relaxation experiment, the parameter τ is determined. Then, one can use the above expression in a regression to find the values of k_{-1} and k_1.